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Oracle Fusion Distributed Order Orchestration – Attachments

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Introduction

Attachment is a supplementary document that is attached to a primary document. For instance, a custom order of furniture has a product spec attached to it. Oracle Fusion Distributed Order Orchestration (DOO) provides end-to-end support for attachments: from receiving attachments in sales order to sending attachments to the order fulfillment system. The implementation of attachments involves mapping attachment attributes from the sales order to the DOO enterprise business message (EBM) for creating a sales order and retrieving attachments in the connector for the order fulfillment system.

End-to-End Support for Attachments

Oracle Fusion Distributed Order Orchestration (DOO) has end-to-end support for attachments: from receiving attachments in sales order to sending attachments to the order fulfillment system. Order manager can also view attachment details in Order Orchestration Work Area.

Receiving Attachments in Sales Order

DOO receives the sales order from the order capture system using a sales order enterprise business object (EBO). DOO then cross-references the order capture system values to Oracle Fusion values to provide a singular representation across disparate systems. An orchestration order is the fulfillment oriented representation of the sales order used within DOO and is created using sales order attributes and enriched through product transformation and fulfillment attribute defaulting.

A sales order may contain one or more than one attachment, where it is associated to either the sales order header or one of the sale order lines or line schedules. The attachments in a sales order are carried over and become part of the orchestration order. The association between the attachment and order or line is retained in the corresponding orchestration order:

- Attachment associated to the sales order header is associated to the orchestration order header
- Attachment associated to the sales order line or line schedule is associated to the corresponding fulfillment line

The attributes of the attachments from the sales order are retained in the orchestration order:

TABLE 1. ATTACHMENT ATTRIBUTES

ATTRIBUTE	DESCRIPTION	COMMENTS
Type	Three types are supported: File, Text, and URL	
Title	The user name of the attachment	
Description	Additional information on the attachment	
Document Content	• If Type is URL, it is the URL entered • If Type is File, it is the file content • If Type is Text, it is the text entered	

DOO receives the change to the attachments in a change order from the order capture system, such as adding or deleting an attachment and modifying an existing attachment. The orchestration order is updated accordingly after DOO accepts the change order.

Viewing Attachments in Order Orchestration Work Area

Order manager can view the attachment details in Order Orchestration Work Area:

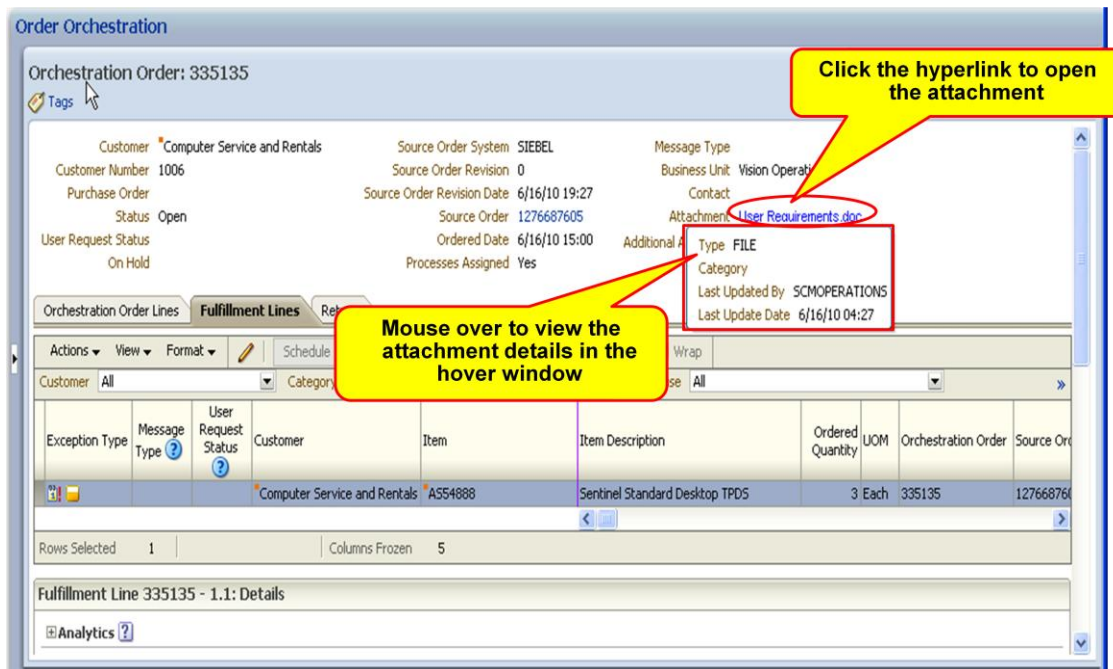


Figure 1. View Attachments

The attachments from the sales order are retained intact in the orchestration order. Order manager is not allowed to modify or delete an attachment in the orchestration order. Adding a new attachment to the orchestration order is not allowed either.

Sending Attachments to Order Fulfillment System

DOO sends a fulfillment request to the order fulfillment system to carry out certain order fulfillment function. Attachments are part of the orchestration order and are accessible to the order fulfillment system through a public web service called Orchestration Order Attachments Service. One of its methods GetAttachment allows the order fulfillment system to retrieve the attachments that are associated to the orchestration order and are relevant for the requested fulfillment function.

Implementing Attachments

The implementation of attachments involves mapping the attachment attributes from a sales order to the DOO enterprise business message (EBM) for creating a sales order in the connector for the order capture system and including the logic of retrieving the attachments and mapping the attachment attributes in the connector for the order fulfillment system.

Mapping Attachment Attributes in Connector for Order Capture System

DOO communicates with the order capture system using the sales order enterprise business object. Attachment is a component of the sales order and is defined as following:

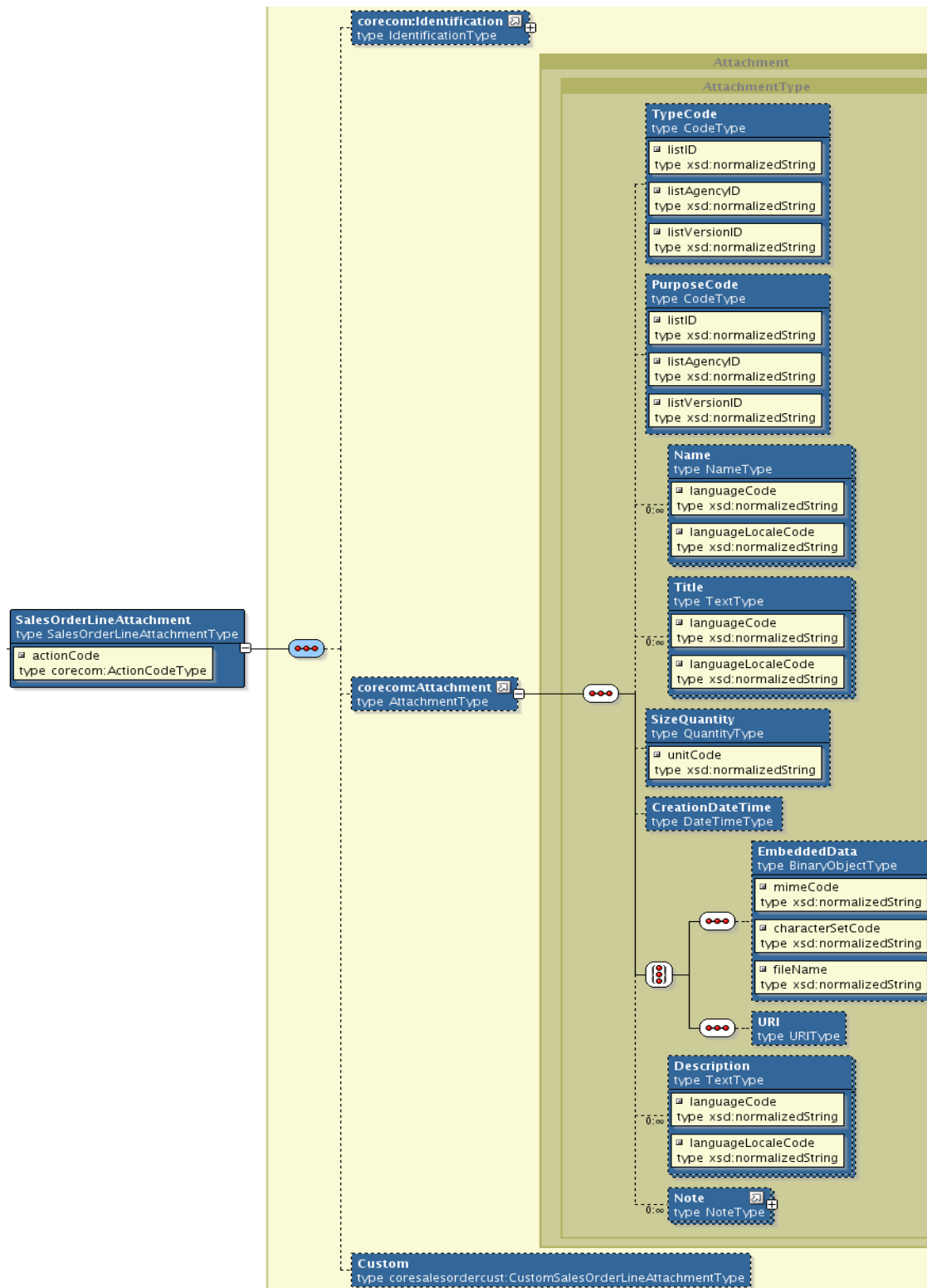


Figure 2. Attachment

The enterprise business message defines the elements that are used as messages in service options. It's an application-agnostic encapsulation of an enterprise business object. To submit a new sales order to DOO, you must transform the sales order (including the attachment attributes) into the DOO enterprise business message for creating a sales order before invoking the public web service SalesOrderOrchestrationService and relevant operation in the connector for the order capture system. The same DOO enterprise business message is also used for changing or cancelling a sales order. The following figure illustrates the mapping from a sales order XSD (XML Schema Definition) to the ProcessSalesOrderFulfillmentEBM (DOO EBM for creating/changing/cancelling a sales order):

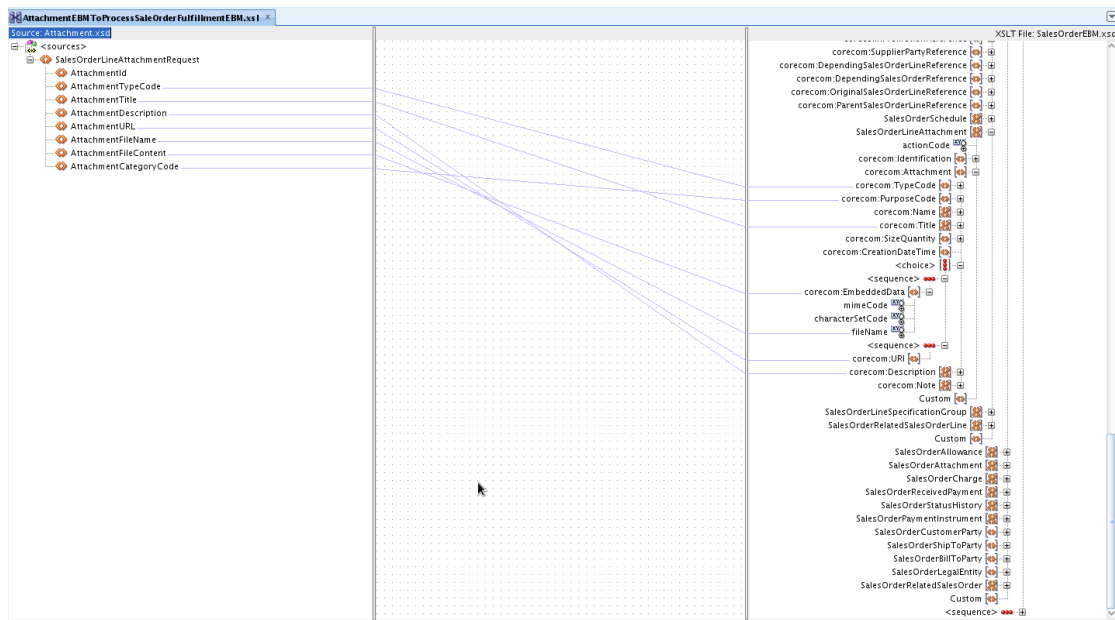


Figure 3. Map Sales Order Attachment Attributes to ProcessSalesOrderFulfillmentEBM

The attachment attributes in the processSalesOrderFulfillmentEBM are under SalesOrderAttachment (for attachments associated to the sales order header), SalesOrderLineAttachment (for attachments associated to the sales order line), or SalesOrderLineScheduleAttachment (for attachments associated to the sales order line schedule):

TABLE 2. ATTACHMENT ATTRIBUTES IN PROCESSSALESORDERFULFILLMENTEBM

ATTRIBUTE	DESCRIPTION
coreoom:Attachment/coreoom:TypeCode	Three types are supported: File, Text, and URL
coreoom:Attachment/coreoom:Title	The user name of the attachment
coreoom:Attachment/coreoom:Description	Additional information on the attachment
coreoom:Attachment/coreoom:URI	If Type is URL, it is the URL entered

coreoom:Attachment/coreoom:EmbeddedData/FileName	If Type is File, it is the name of the file
coreoom:Attachment/coreoom:EmbeddedData	• If Type is File, it is the file content • If Type is Text, it is the text entered Note: If the attachment content is not included in the sales order, you need to retrieve the attachment content before transforming the sales order into the DOO EBM

DOO doesn't send the attachments to the order capture system. Therefore, there is no need to map the attachment attributes when transforming the response received from the DOO service to a format that the order capture system understands.

Retrieving Attachments and Mapping Attachment Attributes in Connector for Order Fulfillment System

DOO sends a fulfillment request to the order fulfillment system to carry out certain order fulfillment function. If attachments are relevant for the requested fulfillment function, you can retrieve the attachments by invoking the GetAttachment method of the Orchestration Order Attachments Service in the connector for the order fulfillment system. Orchestration Order Attachment Service is a public web service. For service path, abstract WSDL URL and other details, see Orchestration Order Attachment Service in [Oracle Enterprise Repository](#) (OER).

The following are the input parameters for the GetAttachment method:

TABLE 3. ORCHESTRATION ORDER ATTACHMENT SERVICE: GETATTACHMENT METHOD

PARAMETER	DESCRIPTION	COMMENTS
SourceEntityName	The name of the entity that the attachment(s) are associated to	• DOO_HEADERS_ALL for orchestration order header • DOO_FULFILL_LINES_ALL for fulfillment line
SourceTableName	The name of database table for the entity that the attachment(s) are associated to	• DOO_HEADERS_ALL for orchestration order header • DOO_FULFILL_LINES_ALL for fulfillment line
SourcePK1Value	The first identifier of the entity that the attachments(s) are associated to	• HEADER_ID for orchestration order header • FULFILL_LINE_ID for fulfillment line
SourcePK2Value	The second identifier of the entity that the attachment(s) are associated to	Not applicable

The GetAttachment method retrieves all the attachments associated to the given fulfillment line or orchestration order header and returns the OrchestrationAttachmentSDOs as output, which contains all the attachment information:

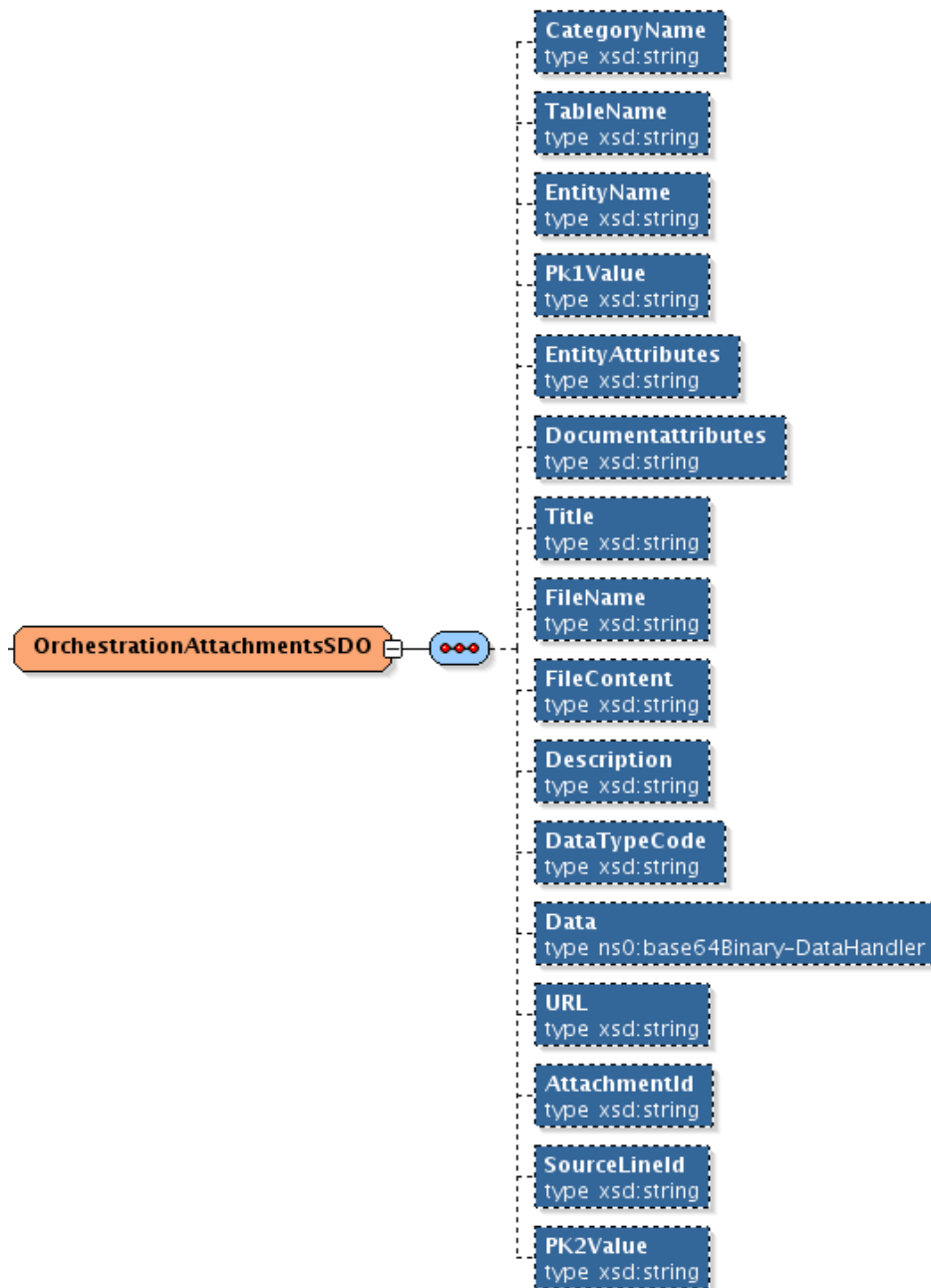


Figure 4. OrchestrationAttachmentSDO

You need to make separate calls in the connector for the order fulfillment system to retrieve the attachments associated to a fulfillment line and the corresponding orchestration order header. Here is the sample code for retrieving the attachments associated to a fulfillment line:

```
<soap:Body>
```

```
<ns1:getAttachment xmlns:ns1="http://xmlns.oracle.com/apps/scm/doo/common/attachments/model/types/">
```

```
<ns1:dooGetParameters xmlns:ns2="http://xmlns.oracle.com/apps/scm/doo/common/attachments/model/">

  <ns2:sourceEntityName>DOO_FULFILL_LINES_ALL</ns2:sourceEntityName>

  <ns2:sourcePK1Value>100000045762 </ns2:sourcePK1Value>

  <ns2:sourceTableName>DOO_FULFILL_LINES_ALL</ns2:sourceTableName>

</ns1:dooGetParameters>

</ns1:getAttachment>

</soap:Body>
```

Figure 5. Retrieve Attachments

Before invoking the order fulfillment system's service and operation designated for the requested fulfillment function, you need to transform the DOO message (including the attachment attributes) into a format that the order fulfillment system understands. This is similar to the transformation that is done in the connector of the order capture system except in the reverse way.

DOO doesn't accept attachments from the order fulfillment system. Therefore, there is no need to map the attachment attributes when transforming the order fulfillment system message into a format that DOO understands for inbound response.



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